1

Introduction

Rydberg atoms, atoms in states of high principal quantum number, n, are atoms with exaggerated properties. While they have only been studied intensely since the nineteen seventies, they have played a role in atomic physics since the beginning of quantitative atomic spectroscopy. Their role in the early days of atomic spectroscopy is described by White.[1]

The first appearance of Rydberg atoms is in the Balmer series of atomic H. Balmer's formula, from 1885, for the wavelengths of the visible series of atomic H, is given by[1]

$$\lambda = \frac{bn^2}{n^2-4}, \tag{1.1}$$

where $b = 3645.6$ Å. We now recognize Eq. (1.1) as giving the wavelengths of the Balmer series of transitions from the $n = 2$ states to higher lying levels.

While the H atom was the first atom to be understood in a quantitative way, other atoms played a crucial role in unravelling the mysteries of atomic spectroscopy. For example, Liveing and Dewar[2] made the important observation that the observed spectral lines of Na could be grouped into different series. Specifically they observed the Na ns–3p and nd–3p emissions from an arc. Up to the 9s and 8d states they observed doublets due to the fine structure splitting of the 3p state, but for the 10s and 9d states they were unable to resolve the 3p doublet. The ns–3p doublets appeared as sharp lines while the nd–3p doublets appeared as diffuse lines. Most important, they recognized that these sharp and diffuse doublets were members of two series of related lines, although they were unsuccessful in their attempts to relate the wavelengths of the observed transitions.

Knowing, as we do today, that the Na d states are almost degenerate with the higher angular momentum states of the same n, we are not surprised that the series of transitions to the d Rydberg series is diffuse. They are much more likely to suffer pressure broadening. In contrast to Na, Liveing and Dewar noted that in the K spectrum the s and d series were not particularly different, both being "more or less diffuse".[2] We now realize that the difference between Na and K is due to the fact that both the s and the d Rydberg states of K are energetically well removed from the high angular momentum states.

A crucial advance was made by Hartley, who in his studies of the spectra of Mg, Zn, and Cd, first realized the significance of the frequency of a transition as opposed to the experimentally measured wavelength.[1,3] Hartley observed that the splittings of series of multiplets always had the same wavenumber splitting, irrespective of the wavelengths of the transitions. The wavenumber, ν, is the inverse of the wavelength in vacuum. The importance of this realization is hard to overstate and is immediately apparent if we rewrite Balmer's formula in terms of the wavenumber of the observed lines instead of the wavelength,

$$\nu = \left(\frac{1}{4b}\right)\left(\frac{1}{4} - \frac{1}{n^2}\right). \tag{1.2}$$

It is evident that it simply reflects the energy difference between the $n = 2$ and higher lying states.

Following the precedent of Liveing and Dewar, Rydberg began to classify the spectra of other atoms, notably alkali atoms, into sharp, principal, and diffuse series of lines.[4] Each series of lines has a common lower level and a series of ns, np, or nd levels, respectively, as the upper levels of the sharp, principal, and diffuse series. He realized that the wavenumbers of the series members were related and that the wavenumbers of the observed lines could be expressed as[1,4]

$$\begin{aligned}\nu_s &= \nu_{\infty s} - \frac{Ry}{(n-\delta_s)^2},\\ \nu_p &= \nu_{\infty p} - \frac{Ry}{(n-\delta_p)^2},\\ \nu_d &= \nu_{\infty d} - \frac{Ry}{(n-\delta_d)^2},\end{aligned} \tag{1.3}$$

where the constants $\nu_{\infty s}$, $\nu_{\infty p}$, and $\nu_{\infty d}$ and δ_s, δ_p, and δ_d are the series limits and quantum defects of the sharp, principal, and diffuse series. The constant Ry is a universal constant, which can be used to describe the wavenumbers of the transitions, not only for different series of the same atom, but for different atoms as well. This constant, Ry, is now known as the Rydberg constant. Defining the wavenumber as the inverse of the wavelength in air, Rydberg[4] assigned Ry the value 109721.6 cm^{-1}. Recognition of the generality of the Rydberg constant is one of Rydberg's two major accomplishments. The other was to show that all the spectral lines from an atom are related. Specifically, he realized that $\nu_{\infty s}$ and $\nu_{\infty d}$ were the same to within experimental error. For example, for the series originating in the Na $3p_{3/2}$ and $3p_{1/2}$ levels he assigned the values 24485.9 and 24500.5 cm^{-1} for $\nu_{\infty s}$, and 24481.8 and 24496.4 cm^{-1} for $\nu_{\infty d}$. In addition, he observed that $\nu_{\infty p} - \nu_{\infty s}$ was equal to the wavenumber of the first member of the principal series. He reasoned that it should be possible to see other series which did not have the first term of the principal series in common, but some other term instead. This notion led him to a general expression for the wavenumbers of lines connecting different series. The wavenumbers of lines connecting the s and p series, for example, are given by[4]

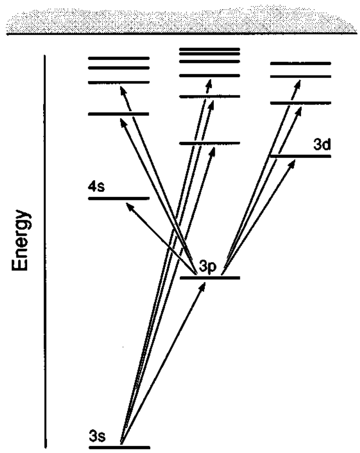

Fig. 1.1 Energy level diagram for Na showing the sharp 3p–*n*s, principal 3s–*n*p, and diffuse 3p–*n*d series.

$$\pm\nu = \frac{Ry}{(m-\delta_s)^2} - \frac{Ry}{(n-\delta_p)^2}. \tag{1.4}$$

In Eq. (1.4) the + sign and constant n describe a sharp series of s states and the minus sign and a constant m describe a principal series of p states. If we consider the special case $\delta_s = \delta_p = 0$ and $m = 2$ we arrive at Balmer's formula for the H transitions from n=2.

From the early spectroscopic work it is possible to construct an energy level diagram for Na, as shown in Fig. 1.1. From this figure it is apparent that the difference in the principal and sharp series limits is the wavenumber of the 3s–3p transition. It is also apparent that the Rydberg states, states of high principal quantum number n, lie close to the series limit.

The physical significance of high n did not become clear until Bohr proposed his model of the H atom in 1913.[1] In spite of its shortcomings, the Bohr atom contains the properties of Rydberg atoms which make them interesting. The basis of the Bohr atom is an electron moving classically in a circular orbit around an ionic core. To the currently accepted ideas of classical physics Bohr added two notions; that the angular momentum was quantized in integral units of $\hbar$, Planck's constant divided by 2π, and that the electron did not radiate continuously in a classical manner, but gave off radiation only in making transitions between states of well defined energies. The first was an assumption, and the second was forced upon him by the experimental observations.

The Bohr theory can be summarized as follows. An electron of charge $-e$ and mass m in a circular orbit of radius r about an infinitely heavy positive charge of Ze obeys Newton's law for uniform circular motion[5]

$$\frac{mv^2}{r} = \frac{kZe^2}{r^2}, \tag{1.5}$$

where $k = 1/4\pi\varepsilon_0$, ε_0 being the permittivity of free space. Adding Bohr's requirement of the quantization of angular momentum yields

$$mvr = n\hbar. \tag{1.6}$$

Combining Eqs. (1.5) and (1.6) leads immediately to an expression for the radius r of the orbit

$$r = \frac{n^2\hbar^2}{Ze^2mk}. \tag{1.7}$$

In other words, the size of the orbit increases as the square of the principal quantum number, and states of high n have very large orbits. The energy, W, of a state is obtained by adding its kinetic and potential energies,

$$W = \frac{mv^2}{2} - \frac{kZe^2}{r} = \frac{-k^2Z^2e^4m}{2n^2\hbar^2}. \tag{1.8}$$

The energies are negative, i.e. the electron is bound to the proton, and the binding energy decreases as $1/n^2$. The allowed transition frequencies are the differences in the energies. Explicitly

$$W_2 - W_1 = \frac{k^2Z^2e^4m}{2\hbar^2}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right). \tag{1.9}$$

Comparing this expression to the general form of Rydberg's formula we can see that

$$Ry = \frac{k^2Z^2e^4m}{2\hbar^2}. \tag{1.10}$$

Historically, the most important aspect of the Bohr atom was that it related the spectroscopic Rydberg constant to the mass and charge of the electron. From our present point of view, the Bohr atom physically defines Rydberg atoms and shows us why they are interesting. In high n, or Rydberg, states the valence electrons have binding energies which decrease as $1/n^2$ and orbital radii which increase as n^2. In other words, in a Rydberg atom the valence electron is in a large, loosely bound orbit. Although Rydberg atoms of principal quantum number well over 100 have been studied in the laboratory, it is instructive to consider a relatively low Rydberg state, $n = 10$, and compare it to a ground state atom. The ground state of H is bound by 1 Ry, 13.6 eV, and has an orbital radius of $1a_0$ and a geometric cross section of a_0^2. In contrast, the $n = 10$ state has a binding energy of 0.01 Ry, an

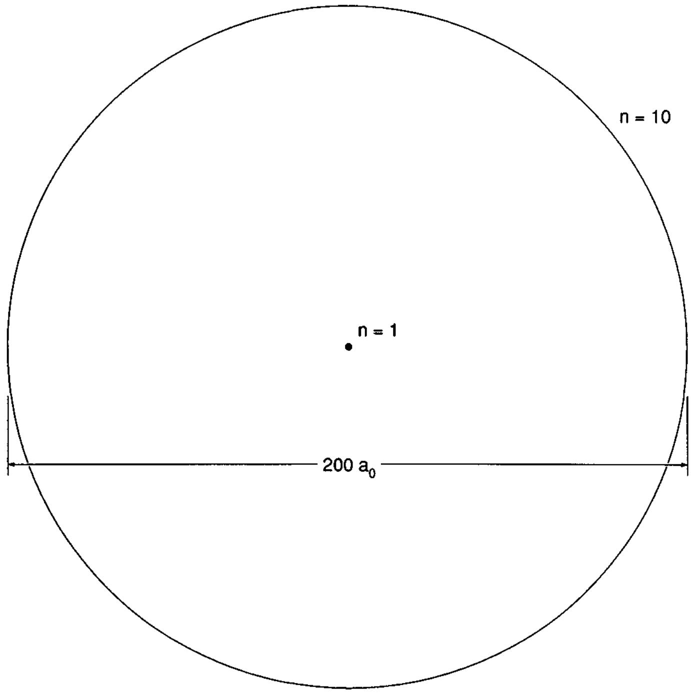

Fig. 1.2 Bohr orbits of $n = 1$ and $n = 10$. The solid dark circle at the center is the $n = 1$ orbit of radius $1a_0$. The $n = 10$ orbit has radius $100a_0$. $1a_0 = 0.53$ Å.

orbital radius of $100a_0$, and a geometric cross section of $10^4 a_0{}^2$. Fig. 1.2 is a drawing of the Bohr orbits for $n = 1$ and $n = 10$, which demonstrates graphically the difference between a Rydberg atom and a ground state atom. The binding energy of the $n = 10$ atom is comparable to thermal energies, and the geometric cross section is orders of magnitude larger than gas kinetic cross sections.

Once the Bohr theory was formulated it was apparent that Rydberg atoms should have bizarre properties. Why then were Rydberg atoms not studied more extensively until the nineteen seventies? There are two reasons. First, the most pressing problem at the time was not the properties of atoms near the series limit, but the development of the quantum theory. Second, there was simply no means of making Rydberg atoms efficiently enough to study them in the detail that is now possible. Nonetheless, the most sophisticated tool available at the time, high resolution absorption spectroscopy, was used in the first experiments to demonstrate the bizarre properties of Rydberg atoms. By examining the fine details of the spectra, the splittings, shifts, and broadenings, it was possible to learn a great deal about Rydberg states.

One of the first properties to be explored was the sensitivity of Rydberg atoms to external electric fields, the Stark effect. While ground state atoms are nearly immune to electric fields, relatively modest electric fields not only perturb the Rydberg energy levels, but even ionize Rydberg atoms, as was shown in early

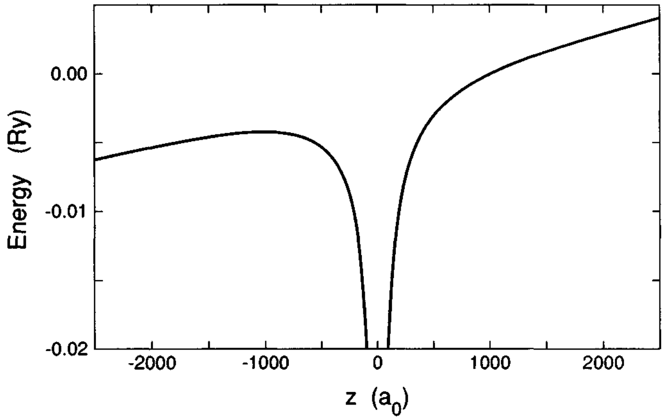

Fig. 1.3 Combined Coulomb–Stark potential along the z axis for a field of 5.14 kV/cm in the z direction.

studies of the Balmer series in electric fields of up to 10^6 V/cm.[6,7] The Balmer lines exhibit a splitting which is approximately linear in the electric field, and the components actually disappear at well defined values of the electric field due to field ionization of the Rydberg state.

While the Bohr atom is of no help in understanding the splittings of the Balmer lines, using it we can calculate the field at which a state is ionized by an electric field. Consider a H atom with its nucleus at the origin in the presence of an electric field in the z direction. The potential experienced by an electron moving along the z axis is given by

$$V = -\frac{k}{|z|} + Ez \tag{1.11}$$

and is shown in Fig. 1.3. Only electrons with energies lower than the local maximum of the potential, at $z = -1000a_0$, are classically bound. In the real three dimensional potential the local maximum at $z = -1000a_0$ is a saddle point, and electrons with energies above the saddle point of the potential are ionized by the field. The potential at the saddle point is given by

$$V_s = -2\sqrt{kE}. \tag{1.12}$$

If this energy is equated to the energy of the Rydberg state, $-Ry/n^2$, we find the classical field for ionization of a state of principal quantum number n

$$E_c = \frac{Ry^2}{4kn^4}. \tag{1.13}$$

This formula is usually encountered in atomic units, in which it reads

$$E_c = 1/16n^4. \tag{1.14}$$

In the next chapter we shall introduce atomic units. Although we have derived Eq. (1.13) with no regard for tunneling or the shifts of the energy levels in an electric field, it is quite a useful expression, as it is, in a practical sense, nearly exact for nonhydrogenic atoms.

The size of a Rydberg atom is large, the geometric cross section scaling as n^4, and, not surprisingly, early experiments were focused on this aspect of Rydberg atoms. A classic experiment was undertaken by Amaldi and Segre to observe the energy shifts of the high lying members of the Rydberg series of K in the presence of high pressures of rare gas atoms.[8] Specifically, they observed the shifts of the 4s–np absorption lines due to the added rare gas. The expectation at the time was that a red shift of the lines would be observed due to the fact that the space between the Rydberg electron and the K^+ ion is filled with a polarizable dielectric, the rare gas atoms, instead of vacuum. While the pressure shifts for Ar and He were to the red, the pressure shift for Ne was to the blue, an observation completely incompatible with the dielectric model. Shortly after the experiment Fermi explained the pressure shifts as arising predominantly from the short range scattering of the Rydberg electron from the rare gas atom, not from the anticipated dielectric effect.[9] This experiment was perhaps the first of many in which the result was diametrically opposed to expectations.

Due to their large size, Rydberg atoms also exhibit large diamagnetic energy shifts. Since the Zeeman and Paschen–Back effects are proportional to the angular momentum, they are not particularly striking in the optically accessible low ℓ Rydberg states. The diamagnetic shifts on the other hand depend on the area of the orbit and thus scale as n^4. The diamagnetic shifts are very difficult to see in low n states, but in high n states they are quite evident. The diamagnetic shifts were first observed by Jenkins and Segre in the absorption spectrum of K taken in the 27 kG field of the Berkeley cyclotron.[10] As n increases the field first mixes states of different ℓ, then mixes states of different n. Only within the last 20 years have experiments progressed substantially beyond these first experiments of Jenkins and Segre. Ironically, the study of Rydberg atoms in magnetic fields, one of the first topics studied, is still a topic of active research 50 years later.

The interest in Rydberg atoms waned until it became apparent that they might play an important role in real physical systems. In particular it became apparent that they were important astrophysically.[11] The radio recombination lines detected by radio astronomers are the emissions from transitions between Rydberg states subsequent to radiative recombination. A particularly favorable frequency regime in which to search for radio signals is 2.4 GHz, which corresponds to the $n = 109$ to $n = 108$ transition. While it is not surprising that Rydberg atoms might be found in interstellar space, where the density is very low, they also play an important role in astrophysical and laboratory plasmas.[12,13] Radiative recombination of low energy electrons and ions results in Rydberg atoms, and dielectronic recombination of higher energy electrons and ions often proceeds by the capture of the electrons into autoionizing Rydberg states and the subsequent radiative decay to bound Rydberg states. Furthermore, the microscopic electric fields in a plasma cut off the Rydberg series at or near the classical ionization limit, as described by Eq. (1.4). Precisely how the Rydberg series is terminated is a matter of some importance, for it determines the thermodynamic properties of the plasma.[12]

The single event which probably contributed the most to the resurgence of interest in Rydberg atoms was the development of the tunable dye laser.[14,15] With a tunable dye laser and very simple apparatus it became possible to excite large numbers of atoms to a single, well defined Rydberg state. This ability allowed one to study a wide range of properties in astonishing detail. Many properties familiar from low lying excited states, such as the radiative lifetimes, energy level spacings, and collision cross sections were measured systematically, and in many, but not all, cases the properties followed the expected extrapolations from the low lying states. Perhaps more interesting were the investigations of the properties which are in a practical sense peculiar to Rydberg atoms. A good example of such properties is the effect of an electric field. While the gross effects of electric fields on Rydberg atoms had been observed long before in optical spectra, and worked out theoretically for some cases, they had not been understood in a systematic way, mainly because there appeared to be no pressing reason to do so. Not only do electric fields produce dramatic effects on Rydberg atoms, but they also afford us novel techniques for the detection and manipulation of Rydberg atoms. To exploit fully these techniques has required a systematic study of the static and dynamic effects of electric fields.

The most fascinating opportunities presented by Rydberg atoms are those which take advantage of their exaggerated properties to carry out experiments that would be impossible in other systems. One example of this is the interaction of atoms with radiation fields. In the low intensity limit, Rydberg atoms provide an ideal system with which to test the interaction of an atom with the vacuum, particularly with the structured vacuum produced by a resonant cavity. Outstanding examples are the one atom maser[16] and the two photon maser.[17] Rydberg atoms have two properties which make them ideal for such investigations. First, the characteristic frequencies of Rydberg atom transitions are low, and the wavelengths correspondingly long, making the construction of resonant or near resonant cavities a straightforward matter. Second, the dipole moments of Rydberg atoms are so large that they have usable radiative decay rates in spite of the low transition frequencies. At the other intensity extreme, advances in laser technology have made atoms in strong radiation fields, fields comparable in magnitude to the coulomb field, a topic of both practical and fundamental interest.[18–20] While it is apparent that any practical applications will involve intense lasers and ground state atoms and molecules, a Rydberg atom in a microwave field is an ideal system in which to carry out quantitative measurements of the properties of atoms in strong radiation fields.[21]

All the properties we have discussed thus far are those of atoms with one Rydberg electron. Double Rydberg atoms, atoms in which there are two highly excited electrons, exhibit pronounced correlation between the motions of the two electrons and have properties which differ in a qualitative way from those of normal Rydberg atoms.[22] While some experiments have been done, they have only scratched the surface, and much is yet to be learned about these exotic atoms.

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